

Name:Uvasri.S

Register Number: 250701801 Experiment Name:K-Means Clustering

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
df=pd.read_csv('Mall_Customers.csv')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Age	200 non-null	int64
2	Annual Income (k\$)	200 non-null	int64
3	Spending Score (1-100)	200 non-null	int64

```
dtypes: int64(4), object(1)
```

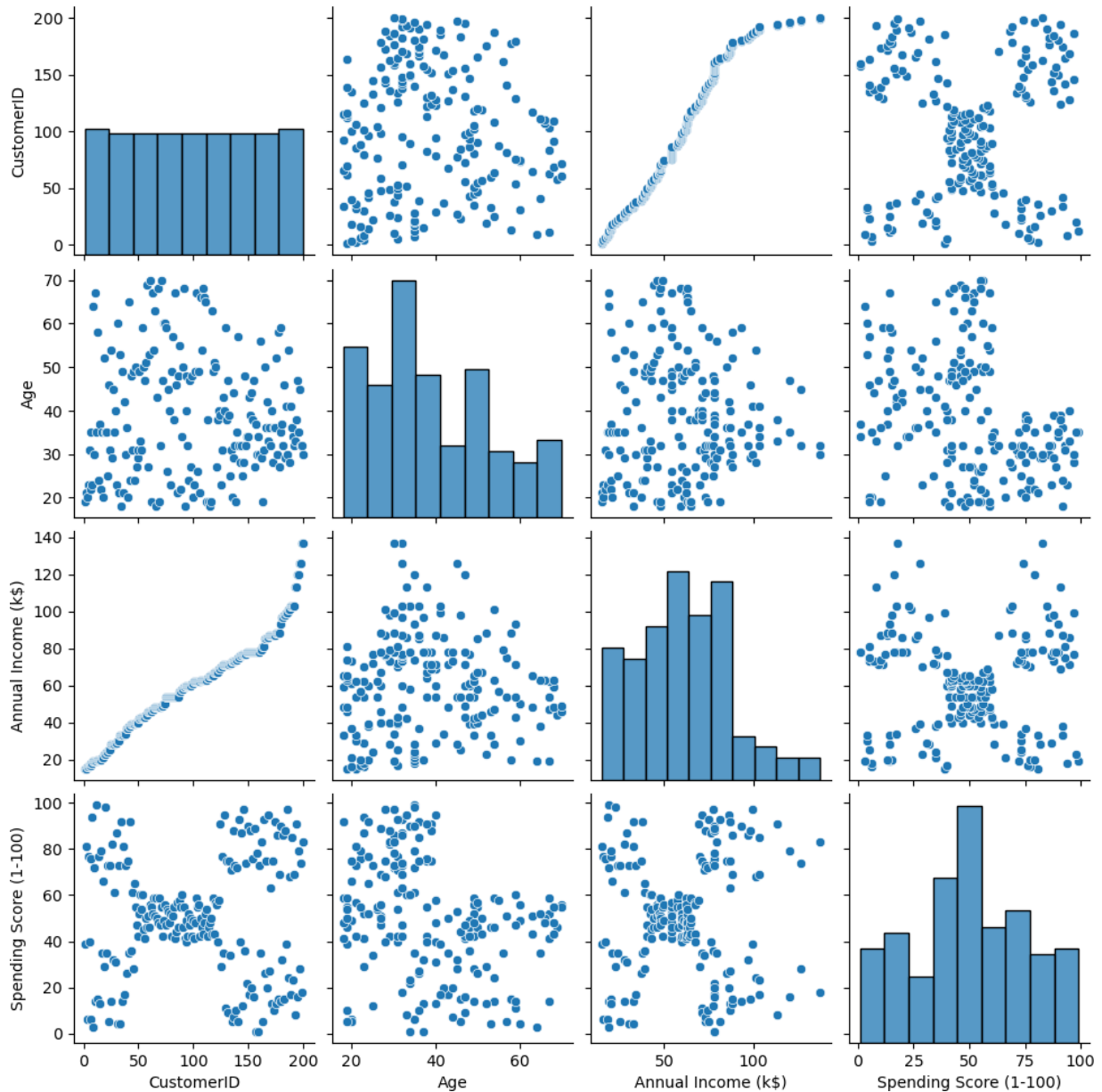
```
memory usage: 7.9+ KB
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
sns.pairplot(df)
```

```
<seaborn.axisgrid.PairGrid at 0x1f67ce53230>
```



```
features=df.iloc[:,[3,4]].values
```

```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

C:\Users\swath\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable

```
OMP_NUM_THREADS=1.
```

```
warnings.warn(
```

```
KMeans(n_clusters=5)
```

```
Final=df.iloc[:,[3,4]]
```

```
Final['label']=model.predict(features)
```

```
Final.head()
```

```
C:\Users\swath\AppData\Local\Temp\ipykernel_5828\470183701.py:2:
```

```
SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame.
```

```
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation:
```

```
https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#
```

```
returning-a-view-versus-a-copy
```

```
Final['label']=model.predict(features)
```

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	3
2	16	6	4
3	16	77	3
4	17	40	4

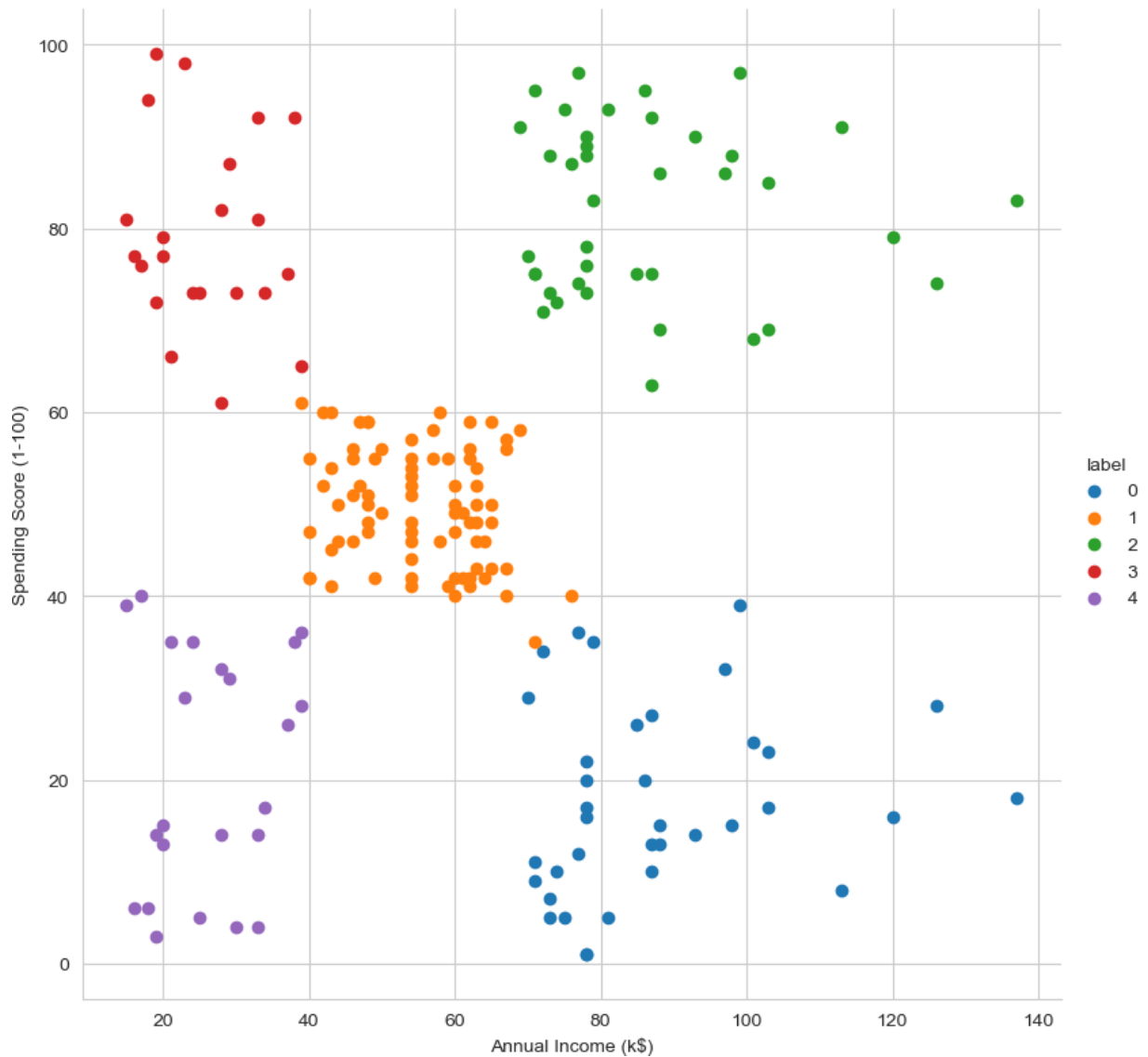
```
sns.set_style("whitegrid")
```

```
sns.FacetGrid(Final,hue="label",height=8) \
```

```
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
```

```
.add_legend();
```

```
plt.show()
```



```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

Cell In[11], line 5

```
model=KMeans(n_clusters=i)
^
```

IndentationError: expected an indented block after 'for' statement on line 4

